Procedure for Model Ice Production and Property Testing at Uni. Melbourne

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1. Model Ice Generation

a. Pre-Cooling

Before the spraying is started the tank and the water need to be pre-cooled. The pre-cooling should be conducted in two steeps. First the tank should be kept at low temperature close to the freezing point of the water – approximately 12h – 24h. This is required to cool the water of a large depth. As warm water rises during the cooling of the surface "heat flux" occurs. If the gradient is too large (which however is an infrastructure specific definition) the warm water rises during the ice growth and might slow down growths or later weaken the ice at the bottom.

In the pre-cooling period before the spraying the temperature should be between -8°C and -12°C for approximately 1h. At this stage "natural ice" will form at the surface and the water surface will be very cold. This is essential to form ice nuclei (by sprayed droplets) from which the later ice will grow.

b. Spraying and Nuclei Seeding

Right before this stage "natural ice" has formed, which needs to be removed before the spraying is initiated. At this stage the water has a super-cooled surface (below freezing point).

At this stage the ambient temperature is still -8°C to -12°C.

A fine water mist is generated by spraying water onto the super-cooled water surface. The water droplets are either fine enough and long enough exposed to the cold air that their freeze before hitting the water surface or the droplets freeze when hitting the water surface.
It is possible to spray warm water as this freezes faster than cold water (Mpemba Effect, see Link). The purpose is to generate a fine grained ice matrix and therefore the ice nuclei need to be seeded close together so that their lateral growth is significantly limited by the neighboring nuclei.

c. Ice Growth

A distinction is made between fine grained ice and columnar ice. In both cases the ambient temperature is not changed and remains between -8°C and -12°C.

Columnar Ice

The growth process is similar to sea ice. After the nuclei are formed and lateral growth ended the crystals grow downwards. The orientation of the crystals is rather random, but there is one direction for the best heat transfer and the best growth. Then a process takes place which called ***geometric selection***, which is ***survival of the fittest***. Those crystals with the best orientation for growth (downwards) are the strongest and cut off the weak ones and grow downward into the water. For this reason the top-layer has a random orientation and after the geometric selection the columnar structure forms.

The growth rate of the columnar ice depends on thickness, temperature and cooling system. It is around 1 mm – 2 mm / hour.

Fine grained ice

Fine grained ice is generated by repeating the spraying process over and over again. This forms a matrix of several crystal layers. In contrary to the columnar ice the fine grained ice “grows upwards”, as ice layer per ice layer is build up upon another. The growth of the crystals is hampered by the lack of water to grow on and the close vicinity of other crystals in all spatial directions. Therefore, the crystals connect with one another via grain boundaries and a fine grained matrix remains.

If the ambient temperature is too cold for too long it can occur that new ice crystals grow from the bottom of the fine grained ice matrix.

d. Consolidation and Tempering

This period may start after the growth. In the case of **fine grained ice** where the growth is concluded with the spraying process. The consolidation at low temperature is again done at low temperature (-8°C to -12°C). The consolidation is needed to establish strong grain boundaries between the grains. The consolidation period for **columnar ice** usually coincides with the growth process.

Once this is established the ice is often stronger than desired and the strength is adjusted by “warming” or “tempering” the ice at -1°C – 0°C degrees.

2. Ice Property measurements

This is a brief introduction to the most essential parameters to be measured.

a. Flexural strength and strain modulus

The flexural strength is usually measured by cantilever beam tests which are cut in-situ. It is also possible to measure the strength with three point bending tests as stated in the **Guidelines of the ITTC 7.5-02-04-02 Test Methods for Model Ice Properties[1].**

It is also be accounted that the failure of cantilever beam tests is partly initiated by stress concentrations at the transition between beam and ice sheet. In 3-point bending tests failure is also not purely initiated by bending, but also by shear stresses in the location of load application. This means that the resulting bending strength obtained from the 3-point bending test might be of limited comparison with the cantilever beam test, which is the standard. This must be accounted!

Furthermore, when extracting the ice from the water it must be done with outmost care, because model ice has a sensitive structure and any touch might change its constitution!

The strain modulus can be measured form the flexural strength test, but therefore the displacement of the load applicator must be measured. It can be that the strain modulus (in the elastic regime, elastic modulus) is non-linear. It might suffice to determine a representative global strain modulus as reference parameter.

b. Density Measurements

The density measurements can be done with a scale and a container of tank water. The illustration below and the formula for the density ratio between ice and water is extracted from the ITTC guidelines 7.5-02-04-02 (Figure 1).

$$\frac{\rho_i}{\rho_w} = \frac{w_2 - w_1}{w_3 - w_1} \qquad (9)$$

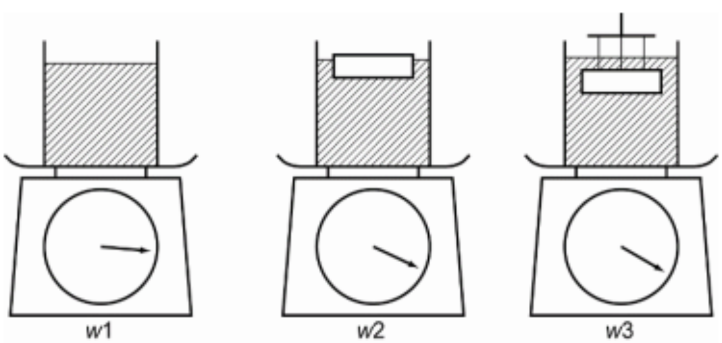

Figure 1: Steps of density measurements (ITTC)

[1] Some of the formulas might contain typos, which are corrected during the current ITTC term, ITTC 29.

c. Ice structure

The ice structure is measured with the approach used for thin sections, where a light source (not monochromatic) is placed behind the ice and two polarized filters on top. The ice needs to be one grain thick (quite thin) to avoid light breaking by two crystals behind each other.

3. Next steps

Here follow a few bullets on what is needed to generate model ice in future respectively wave-ice interaction:

- A small carriage is needed that can move on rails over flume. It needs a pump to spray a fine water mist for seeding ice crystals. The gardening appliances from the home depot cannot generate a mist fine enough. The hand pump container was not so bad
- The flume needs to be water tight
- In the best case the cooling can be set to a program of different temperatures for a certain time
- The tank surface needs to be covered during spraying because the fans generate too much disturbance. This is difficult from a practical point, but possible to do.
- The insulation of the tank needs to be improved so that the heat transfer through the glas is reduced. The panels work ok, but are still not tight enough.
- The wave maker should be modified (use a straight board) and Marco can help with setting it up.
- The strength testing machine needs a loadcell with a lower capacity (10 N, 100N max)